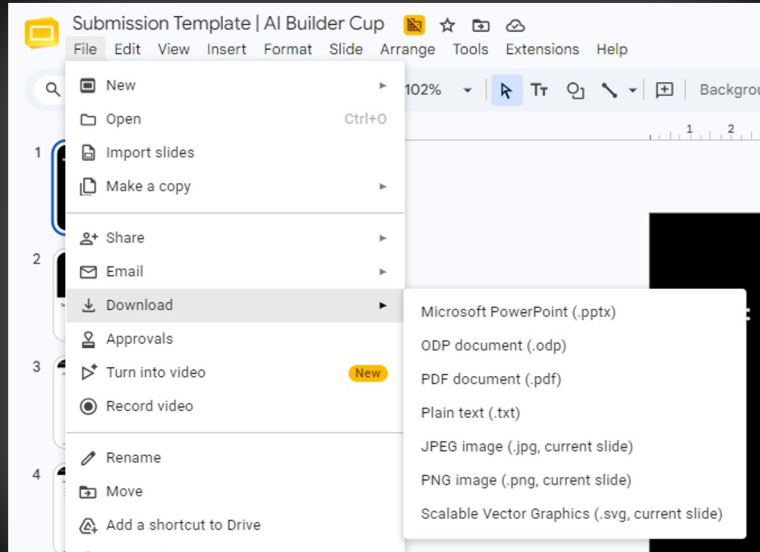




Team Details

- a. Team name: **Green Corridor**
- b. Team leader name: **Nandish**
- c. Problem Statement: **Sustainability & Social Impact — emergency vehicles stuck at red signals; the junction constable is never told**



Note: Please download this template to make your edits and submit the final version.

To download, go to File > Download > Microsoft PowerPoint (.PPTX). Kindly refrain from sending edit requests.

Brief about the idea

Ambulances, fire engines and police vehicles sit at red signals while the officer at the junction has no idea they are coming.

Emergency Green Corridor warns that officer early enough, sized to the queue that is really there, with a spoken alert on their phone; rules decide the order and Gemini extracts and explains.

A crew speaks the patient's condition, one tap confirms the tier, and from then on GPS drives the alerts, the multi-vehicle sequencing and a live hospital handover.

Honest scope: the signals are simulated behind an adapter today; constable alerts are deployable now, on a phone, with no hardware.

Opportunities

- **How different is it from any of the other existing ideas?**

Warning time is sized to the queue, not a fixed radius. Several vehicles are sequenced with shared greens, alerts are spoken, the hospital gets a live handover, and none of it needs hardware.

- **How will it be able to solve the problem?**

The constable clears exactly the queue that is there, minutes before the siren arrives.

- **USP of the proposed solution**

Rules decide; Gemini explains and extracts. Deployable to a constable's phone today, adapter-ready for signal controllers.

List of features offered by the solution

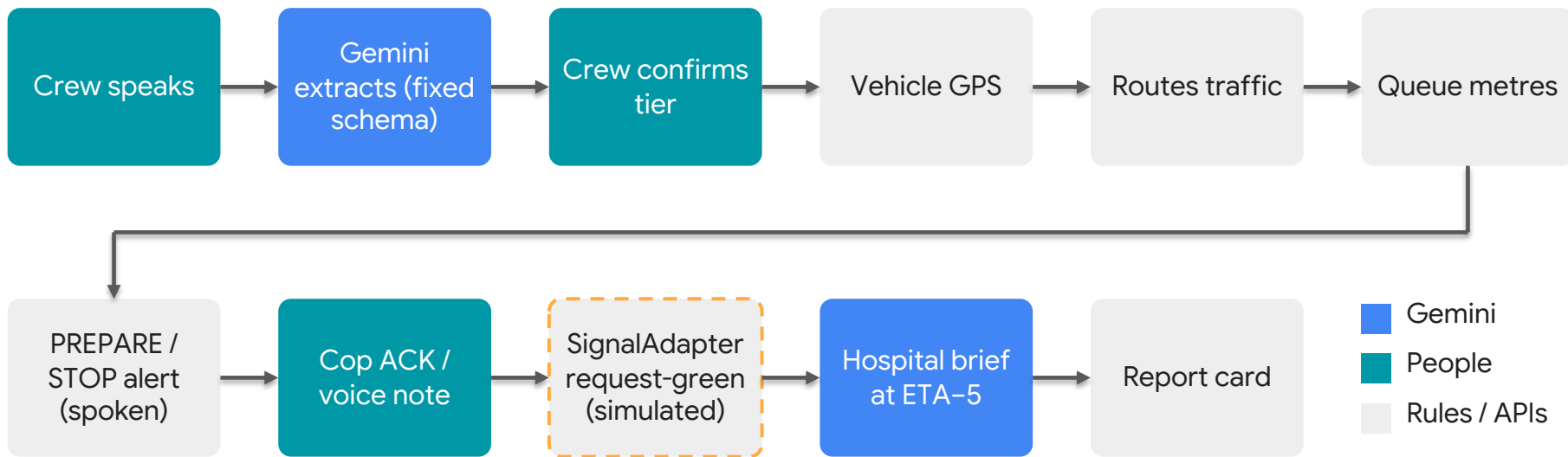
Six screens

- /vehicle: voice or photo triage, tier tap
- /cop: spoken two-stage alert, one ACK
- /hospital: live log, brief, countdown
- /control: runs, junction board, escalations
- /sim: with-vs-without replay, minutes saved
- /dispatch: incident IDs

Engines

- Lead time: queue metres to PREPARE / STOP
- Sequencing and platoon, shared greens
- Cop voice back-channel to rule actions
- Hospital routing agent (ADK) with guard
- ATMIST brief at ETA-5
- After-action report, per-run report cards
- Evaluation harness on voice clips

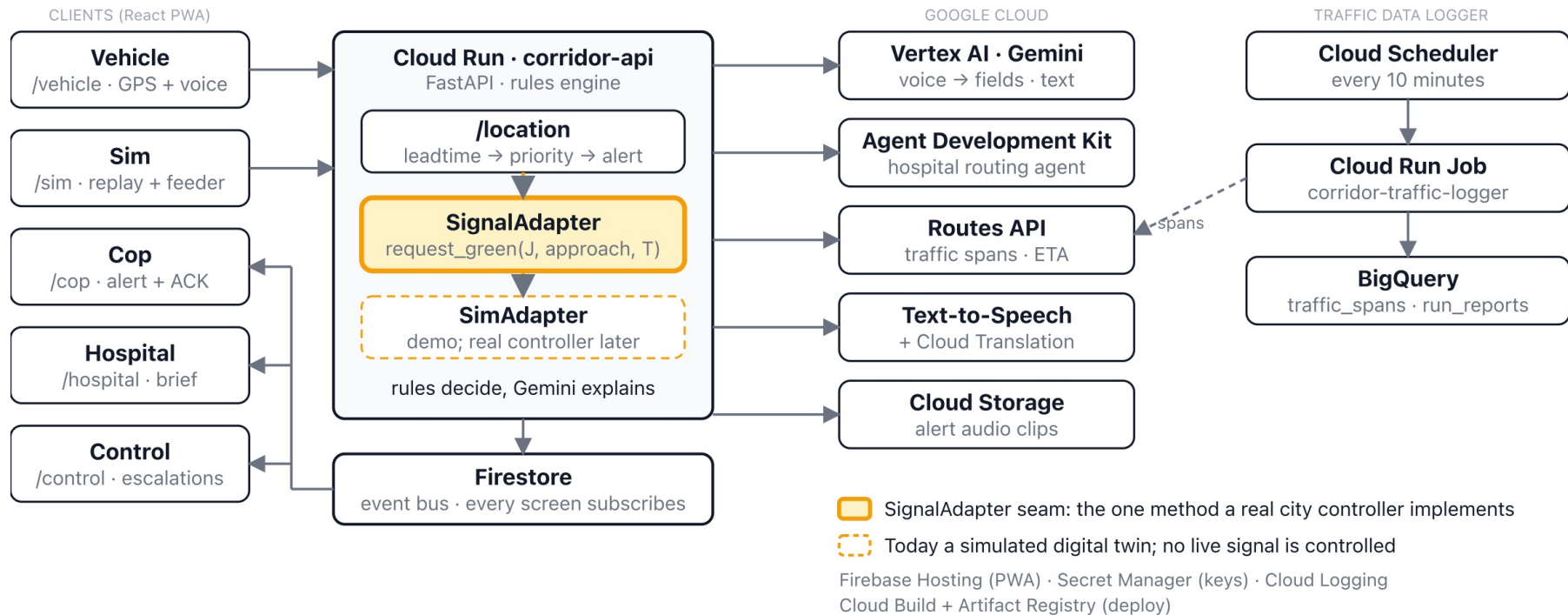
Process flow diagram or Use-case diagram



Rules decide every safety step; Gemini only extracts and explains. Signals are simulated behind SignalAdapter (dashed).

Screens captured from the live rehearsal run: synthetic patients, scripted scenario.

Architecture diagram of the proposed solution



Clients and Firestore event bus around a Cloud Run rules engine; Google services on the right; the SignalAdapter seam is where a real controller plugs in.

Technologies to be used in the solution

Layer	Technology
Generative AI	Gemini on Vertex AI: 3.1 Flash Lite (extraction), 3 Flash (text)
Agents	Agent Development Kit (hospital routing agent)
Compute	Cloud Run, Cloud Run Jobs, Cloud Scheduler
Data	Firestore (event bus), Cloud Storage, BigQuery + BigQuery ML
Hosting	Firebase Hosting (PWA)
Maps	Routes API, Maps JavaScript API
Language	Cloud Text-to-Speech, Cloud Translation
Security and delivery	Secret Manager, Cloud Build, Artifact Registry, Workload Identity Federation
Operations	Cloud Logging, Monitoring, Trace
Frontend / backend	React + Vite / FastAPI (Python 3.12)

Estimated implementation cost (optional)

Measured call budget

- ≈ 3 Routes calls per vehicle-minute
- ~ 17 Gemini + TTS calls per full run
- Cloud Run scales to zero; AI side effects run off the request path

Pilot estimate per vehicle-hour (list prices)

- Routes: $3 \times 60 \times ₹0.85 = ₹153$
- Gemini Flash Lite: $17 \times ₹0.02 = ₹0.34$
- TTS: $\sim 1,000 \text{ chars} \times ₹1.3/1k = ₹1.30$

Total $\approx ₹155$ per vehicle-hour (estimate)

City scale: Routes is $\sim 98\%$ of the bill, so 100 vehicle-hours a day is about ₹15,500 a day; fetching one span per junction and sharing it across vehicles is the lever.

Captured from the live rehearsal run:
synthetic patients, scripted scenario.

Gemini extraction

Prototype Performance report/Benchmarking

Extraction, synthetic voices (12 clips): field 87%, tier 73% (8/11)

Field	Accuracy
age	100%
sex	100%
complaint	36%
conscious	91%
breathing	82%
vitals	100%
trapped_persons	100%

Not field recordings; tier misses are complaint wording.

- Extraction latency: mean 2.9 s per clip
- Rehearsal: J3 alert 4 min out (long queue), J4 alert 1 min out
- 16.5 min saved on the scripted scenario; baseline per junction: cycle/4 + queue/2 m/s (simulated)
- Tests: ~250 Python, 45 web; 98% API coverage
- [load test p50/p95: fill at freeze]

Additional Details/Future Development (if any)

From constable to controller

Today

The alert goes to a person: no city hands a hackathon its signal API. A spoken PREPARE / STOP on a phone, one ACK.

Phase 2

Integrate the city's adaptive signal controllers. The same request-green call already flows through the SignalAdapter seam, so the green fires automatically and constables supervise exceptions.

Phase 3

A learning controller trained on junction data we already log: peak-hour Routes spans to BigQuery.

Also planned: agency rosters, FCM push, production mode, Kannada/Telugu switchable by corridor.

Provide links to your:

- **GitHub Public Repository**
<https://github.com/Nandish3010/ideal-disco>
- **Demo Video Link (3 Minutes)**
[link, 3 minutes — fill]
- **Final Product Link**
<https://green-corridor-2026.web.app>
- **API health**
<https://corridor-api-919512130399.asia-south1.run.app/health>

Google Cloud ~~1125~~

AI  Builder
Cup Build in JAPAC.
Build for the world.

Thank you

